3(a)	acceleration (directly) proportional to displacement	**B1**
	acceleration is in opposite <u>direction</u> to displacement **or** acceleration is (directed) towards a fixed point	**B1**
3(b)(i)	zero	**B1**
3(b)(ii)	E_T is maximum potential energy = mgh $E_T = 94 \times 10^{-3} \times 9.81 \times 0.90 \times 10^{-2}$	**C1**
	$= 8.3 \times 10^{-3}$ J	**A1**
3(b)(iii)	$E_{MAX} = \frac{1}{2} m{v_0}^2$ **and** $v_0 = \omega x_0$ **or** $E_{MAX} = \frac{1}{2}m(\omega x_0)^2$	**C1**
	$8.3 \times 10^{-3} = \frac{1}{2} \times 94 \times 10^{-3} \times \omega^2 \times (12.7 \times 10^{-2})^2$ …leading to $\omega = 3.3$ rad s^{-1}	**A1**
3(c)	$T = 2\pi / \omega$	**C1**
	$2\pi / 3.3 = 2\pi \times (L / 9.81)^{½}$	**C1**
	$L = 0.90$ m	**A1**